

> PRACTICE EXERCISE

- A car completes a journey in 6 h with a speed of 50 km/h. At what speed must it travel to complete the journey in 5 h?
(a) 60 km/h (b) 55 km/h (c) 45 km/h (d) 61 km/h
- Kiran covers a certain distance at 80 km/h and returns back to the same point at 20 km/h. Then, the average speed during the whole journey is
(a) 35 km/h (b) 32 km/h (c) 30 km/h (d) 28 km/h
- Normally Sarita takes 3 h to travel between two stations with a constant speed. One day her speed was reduced by 12 km/h and she took 45 min more to complete the journey. Then, the distance between the two stations is
(a) 60 km (b) 120 km (c) 180 km (d) 95 km
- Rani goes to school at 10 km/h and reaches the school 6 min late. Next day, she covers this distance at 12 km/h and reaches the school 9 min earlier than the scheduled time. What is the distance of her school from her house?
(a) 16 km (b) 12 km (c) 10 km (d) 15 km
- A man travels first 50 km at 25 km/h next 40 km at 20 km/h and then 90 km at 15 km/h. His average speed (in m/s) for the whole journey is
(a) 18 (b) 5 (c) 10 (d) 36
- A boy is running at a speed of p km/h to cover a distance of 1 km. But due to the slippery ground, his speed is reduced by q km/h ($p > q$). If he takes r h to cover the distance, then
(a) $\frac{1}{r} = \frac{pq}{p+q}$ (b) $\frac{1}{r} = p+q$ (c) $r = p-q$ (d) $\frac{1}{r} = p-q$
- A train passes telegraph post in 40 s moving at a rate of 36 km/h. Then, the length of the train is
(a) 400 m (b) 500 m (c) 450 m (d) 395 m
- A person can run around a circular path of radius 21 m in 44 s. In what time will the same person run a distance of 3 km?
(a) 18 min 40 s (b) 16 min 30 s
(c) 18 min 30 s (d) 16 min 40 s
- A car is ahead of a scooter by 30 km. Car goes at the rate of 50 km/h and the scooter goes at the rate of 60 km/h. The scooter overtakes the car after
(a) 3 h (b) 3.5 h (c) 4 h (d) $3\frac{1}{4}$ h
- Two towns A and B are 250 km apart. A bus starts from A to B at 6 : 00 am at a speed of 40 km/h. At the same time another bus starts from B to A at a speed of 60 km/h. The time of their meeting is
(a) 8 : 30 am (b) 8 : 00 am (c) 9 : 00 am (d) 9 : 15 am
- A train T_1 leaves a place P at 5:00 am and reaches another place Q at 9 : 00 am another train T_2 leaves the place Q at 7 : 00 am and reaches the place P at 10 : 30 am. The time at which the two trains cross each other is
(a) 8 : 26 am (b) 7 : 56 am (c) 8 : 15 am (d) 8 : 00 am
- A certain distance is covered at a certain speed. If half of the distance is covered in double time, then the ratio of the two speeds is
(a) 4 : 1 (b) 1 : 4 (c) 2 : 1 (d) 1 : 2
- Two trains start running at the same time from two stations which are 210 km apart and going in opposite directions cross each other at a distance of 100 km from one of the station. The ratio of their speed is
(a) 11 : 9 (b) 10 : 11 (c) 11 : 10 (d) 9 : 11
- A police car is ordered to chase a speeding car that is 5 km ahead. The thief car is travelling at an average speed of 80 km/h and the police car pursues it at an average speed of 100 km/h. How long does it take for the police car to overtake the other car?
(a) 17 min (b) 19 min (c) 13 min (d) 15 min
- Points A and B are 70 km apart on a highway. A car starts from A and another car starts from B at the same time. If they travel in the same direction they meet in 7 h, but they travel towards each other they meet in 1 h. What are the speeds of the cars?
(a) 30 km/h, 40 km/h (b) 36 km/h, 40 km/h
(c) 19 km/h, 20 km/h (d) 40 km/h, 50 km/h
- Assume that the distance that a car runs on 1 L of petrol varies inversely as the square of the speed at which it is driven. It gives a run of 25 km/L at a speed of 30 km/h. At what speed should it be driven to get a run of 36 km/L?
(a) 12.5 km/h (b) 25 km/h (c) 30 km/h (d) 40 km/h
- A man standing on a railway platform observes that a train going in one direction takes 4 s to pass him. Another train of same length going in the opposite direction takes 5 s to pass him. The time taken (in seconds) by the two trains to cross each other will be
(a) $\frac{32}{9}$ (b) $\frac{33}{7}$ (c) $\frac{40}{9}$ (d) $\frac{49}{9}$

- 18.** A bullock cart has to cover a distance of 80 km in 10 h. If it covers half of the journey in $\frac{3}{5}$ th time, what should be its speed to cover the remaining distance in the time left?
(a) 5 km/h (b) 10 km/h (c) 15 km/h (d) 18 km/h
- 19.** By walking at $\frac{3}{4}$ of his usual speed, a man reaches his office 25 min later than usual. His usual time is
(a) 60 min (b) 70 min (c) 75 min (d) 80 min
- 20.** A train covers a distance in 50 min, if it runs at a speed of 48 km/h on an average. The speed at which the train must run to reduce the time of journey to 40 min, will be
(a) 10 km/h (b) 20 km/h (c) 40 km/h (d) 60 km/h
- 21.** Two trains travel in the same direction at 50 km/h and 32 km/h, respectively. A man in the slower train observes that 15 s elapse before the faster train completely passes him. What is the length of the faster train?
(a) 75 m (b) 125 m (c) 150 m (d) $\frac{625}{3}$ m
- 22.** A father and his son start at a point A with speeds of 12 km/h and 18 km/h, respectively and reach another point B. If his son starts 60 min after his father at A and reaches B, 60 min before his father, what is the distance between A and B?
(a) 90 km (b) 72 km (c) 36 km (d) None of these
- 23.** Two trains of lengths 100 m and 150 m are travelling in opposite directions at speeds of 75 km/h and 50 km/h, respectively. What is the time taken by them to cross each other?
(a) 7.4 s (b) 7.2 s (c) 7 s (d) 6.8 s
- 24.** A man can walk uphill at the rate of 2.5 km/h and downhill at the rate of 3.25 km/h. If the total time required to walk a certain distance up the hill and return to the starting position is 4 h 36 min, what is the distance he walked up the hill?
(a) 3.5 km (b) 4.5 km (c) 5.5 km (d) 6.5 km
- 25.** A boat goes 30 km upstream and 44 km downstream in 10 h. In 13 h it can go 40 km upstream and 55 km downstream. The speed of the boat in still water is
(a) 9 km/h (b) 8 km/h (c) 4 km/h (d) 3 km/h
- 26.** A motorboat takes 2 h to travel a distance of 9 km down the current and it takes 6 h to travel the same distance against the current. What is the speed of the boat in still water (in km/h)?
(a) 3 (b) 2 (c) 1.5 (d) 10
- 27.** Ram travels from P to Q at 10 km/h and return at 15 km/h. Shyam travels from P to Q and return at 12.5 km/h. If he takes 12 min less than Ram, then what is the distance between P and Q?
(a) 60 km (b) 45 km (c) 36 km (d) 30 km
- 28.** A person x started 5 min earlier at 60 km/h from a place P, then another person y followed him at 48 km/h, started his journey at 3:05 pm. Which of the following is/are correct.
I. At 3 : 15 pm, x and y are 7 km apart.
II. At 3 : 25 pm, y will overtake x
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 29.** A boy walking at a speed of 20 km/h reaches his school 30 min late. Next time he increases his speed by 4 km/h but still he is late by 10 min.
Which of the following statement is/are correct?
I. The distance of the school from his home is 50 km/h.
II. If he increases his speed by 10 km/h but still late by 10 min, then distance of the school will be 20 km.
(a) Both I and II are correct (b) Only I is correct
(c) Only II is correct (d) Neither I nor II are correct.
- Directions** (Q. Nos.30-31) Ramesh and Prateek start running at same time in opposite directions from two points and after passing each other they complete their journeys in 'x' and 'y' h, respectively.
Then, Speed of Ramesh : Speed of Prateek = $\sqrt{y} : \sqrt{x}$
- 30.** Find the ratio of the speeds of Ramesh and Prateek, if Ramesh and Prateek completed their journeys in 16 h and 25 h after passing each other.
(a) 5 : 4 (b) 5 : 3 (c) 4 : 5 (d) 3 : 5
- 31.** What if two persons completed their journey with speeds in the ratio 7 : 9, then how much time they should have taken to complete their journey after they meet each other.
(a) 81 : 49 (b) $\sqrt{7} : \sqrt{9}$ (c) 49 : 81 (d) 14 : 18

PREVIOUS YEARS' QUESTIONS

- 32.** A train 280 m long is moving at a speed of 60 km/h. What is the time taken by the train to cross a platform 220 m long? ☑ 2012 I
(a) 45 s (b) 40 s (c) 35 s (d) 30 s
- 33.** A car travels along the four sides of a square at speeds v , $2v$, $3v$ and $4v$, respectively. If u is the average speed of the car in its travel around the square, then which one of the following is correct? ☑ 2012 I
(a) $u = 2.25v$ (b) $u = 3v$
(c) $v < u < 2v$ (d) $3v < u < 4v$
- 34.** A car is travelling at a constant rate of 45 km/h. The distance travelled by car from 10 : 40 am to 1 : 00 pm is ☑ 2012 II
(a) 165 km (b) 150 km (c) 120 km (d) 105 km

- 35.** A person travels a certain distance at 3 km/h and reaches 15 min late. If he travels at 4 km/h, he reaches 15 min earlier. The distance he has to travel is **✎ 2013 I**
 (a) 4.5 km (b) 6 km (c) 7.2 km (d) 12 km
- 36.** If a body covers a distance at the rate of x km/h and another equal distance at the rate of y km/h, then the average speed (in km/h) is **✎ 2013 I**
 (a) $\frac{x+y}{2}$ (b) $\sqrt{xy}$ (c) $\frac{2xy}{x+y}$ (d) $\frac{x+y}{xy}$
- 37.** A sailor sails a distance of 48 km along the flow of a river in 8 h. If it takes 12 h to return the same distance, then the speed of the flow of the river is **✎ 2013 I**
 (a) 0.5 km/h (b) 1 km/h (c) 1.5 km/h (d) 2 km/h
- 38.** A train running at the speed of 72 km/h goes past a pole in 15 s. What is the length of the train? **✎ 2013 II**
 (a) 150 m (b) 200 m (c) 300 m (d) 350 m
- 39.** Two cars A and B start simultaneously from a certain place at the speed of 30 km/h and 45 km/h, respectively. The car B reaches the destination 2 h earlier than A . What is the distance between the starting point and destination? **✎ 2013 II**
 (a) 90 km (b) 180 km (c) 270 km (d) 360 km
- 40.** A man cycles with a speed of 10 km/h and reaches his office at 1 : 00 pm. However, when he cycles with a speed of 15 km/h, he reaches his office at 11 : 00 am. At what speed should he cycle, so that he reaches his office at 12 noon? **✎ 2013 II**
 (a) 12.5 km/h (b) 12 km/h (c) 13 km/h (d) 13.5 km/h
- 41.** A train takes 9 s to cross a pole. If the speed of the train is 48 km/h, the length of the train is **✎ 2014 I**
 (a) 150 m (b) 120 m (c) 90 m (d) 80 m
- 42.** A train takes 10 s to cross a pole and 20 s to cross a platform of length 200 m. What is the length of the train? **✎ 2014 II**
 (a) 50 m (b) 100 m (c) 150 m (d) 200 m
- 43.** A train travels at a speed of 40 km/h and another train at a speed of 20 m/s. What is the ratio of speed of the first train to that of the second train? **✎ 2014 II**
 (a) 2 : 1 (b) 5 : 9 (c) 5 : 3 (d) 9 : 5
- 44.** The distance between two points (A and B) is 110 km. X starts running from point A at a speed of 60 km/h and Y starts running from point B at a speed of 40 km/h at the same time. They meet at a point C , somewhere on the line AB . What is the ratio of AC to BC ? **✎ 2014 II**
 (a) 3 : 2 (b) 2 : 3 (c) 3 : 4 (d) 4 : 3
- 45.** A man rides one-third of the distance from A to B at the rate of x km/h and the remainder at the rate of $2y$ km/h. If he had travelled at a uniform rate of $6z$ km/h, then he could have ridden from A to B and back again in the same time. Which one of the following is correct? **✎ 2014 II**
 (a) $z = x + y$ (b) $3z = x + y$
 (c) $\frac{1}{z} = \frac{1}{x} + \frac{1}{y}$ (d) $\frac{1}{2z} = \frac{1}{x} + \frac{1}{y}$
- 46.** In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/h and the time of flight increased by 30 min. The duration of the flight is **✎ 2015 I**
 (a) 1 h (b) 2 h (c) 3 h (d) 4 h
- 47.** With a uniform speed, a car covers a distance in 8 h. Had the speed been increased by 4 km/h, the same distance could have been covered in 7 h and 30 min. What is the distance covered? **✎ 2015 I**
 (a) 420 km (b) 480 km (c) 520 km (d) 640 km
- 48.** A car travels the first one-third of a certain distance with a speed of 10 km/h, the next one-third distance with a speed of 20 km/h and the last one-third distance with a speed of 60 km/h. The average speed of the car for the whole journey is **✎ 2015 I**
 (a) 18 km/h (b) 24 km/h (c) 30 km/h (d) 36 km/h
- 49.** A man rows 32 km downstream and 14 km upstream, and he takes 6 h to cover each distance. What is the speed of the current? **✎ 2015 I**
 (a) 0.5 km/h (b) 1 km/h (c) 1.5 km/h (d) 2 km/h
- 50.** A runs $1\frac{2}{3}$ times as fast as B . If A gives B a start of 80 m, how far must the winning post from the starting point be so that A and B might reach it at the same time? **✎ 2015 I**
 (a) 200 m (b) 300 m (c) 270 m (d) 160 m
- 51.** A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the policeman chases him. The thief and the policeman run at the speed of 10 km/h and 11 km/h, respectively. What is the distance between them after 6 min? **✎ 2015 I**
 (a) 100 m (b) 120 m (c) 150 m (d) 160 m
- 52.** Two persons A and B start simultaneously from two places c km apart and walk in the same direction. If A travels at the rate of p km/h and B travels at the rate of q km/h, then A has travelled before he overtakes B a distance of **✎ 2015 I**
 (a) $\frac{qc}{p+q}$ km (b) $\frac{pc}{p-q}$ km (c) $\frac{qc}{p-q}$ km (d) $\frac{pc}{p+q}$ km

- 53.** By increasing the speed of his car by 15 km/h, a person covers 300 km distance by taking an hour less than before. The original speed of the car was **2015 II**
(a) 45 km/h (b) 50 km/h (c) 60 km/h (d) 75 km/h
- 54.** Two trains, one is of 121 m in length at the speed of 40 km/h and the other is of 99 m in length at the speed of 32 km/h are running in opposite directions. In how much time will they be completely clear from each other from the moment they meet? **2015 II**
(a) 10 s (b) 11 s (c) 16 s (d) 21 s
- 55.** The speeds of three buses are in the ratio 2 : 3 : 4. The time taken by these buses to travel the same distance will be in the ratio. **2015 II**
(a) 2 : 3 : 4 (b) 4 : 3 : 2
(c) 4 : 3 : 6 (d) 6 : 4 : 3
- 56.** Two trains are moving in the same direction at 1.5 km/min and 60 km/h, respectively. A man in the faster train observes that it takes 27 s to cross the slower train. The length of the slower train is **2015 II**
(a) 225 m (b) 230 m (c) 240 m (d) 250 m
- 57.** In a race of 100 m, A beats B by 4 m and A beats C by 2 m. By how many metres (approximately) would C beat B in another 100 m race assuming C and B run with their respective speeds as in the earlier race? **2015 II**
(a) 2 (b) 2.04 (c) 2.08 (d) 3.2
- 58.** Three athletes run a 4 km race. Their speeds are in the ratio 16 : 15 : 11. When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third position is **2015 II**
(a) 1000 m (b) 800 m (c) 750 m (d) 600 m
- 59.** A motorboat, whose speed is 15 km/h in still water goes 30 km downstream and comes back in a total of 4h and 30 min. The speed of the stream is **2015 II**
(a) 4 km/h (b) 5 km/h
(c) 6 km/h (d) 10 km/h
- 60.** A bike consumes 20 mL of petrol per kilometre, if it is driven at a speed in the range of 25-50 km/h and consumes 40 mL of petrol per kilometre at any other speed. How much petrol is consumed by the bike in travelling a distance of 50 km, if the bike is driven at a speed of 40 km/h for the first 10 km, at a speed of 60 km/h for the next 30 km and at a speed of 30 km/h for the last 10 km? **2015 II**
(a) 1 L (b) 1.2 L (c) 1.4 L (d) 1.6 L
- 61.** A passenger train takes 1 h less for a journey of 120 km, if its speed is increased by 10 km/h from its usual speed. What is its usual speed? **2016 I**
(a) 50 km/h (b) 40 km/h (c) 35 km/h (d) 30 km/h
- 62.** A man walking at 5 km/h noticed that a 225 m long train coming in the opposite direction crossed him in 9 s. The speed of the train is **2016 I**
(a) 75 km/h (b) 80 km/h (c) 85 km/h (d) 90 km/h
- 63.** A cyclist moves non-stop from A to B, a distance of 14 km, at a certain average speed. If his average speed reduces by 1 km/h, then he takes 20 min more to cover the same distance. The original average speed of the cyclist is **2016 I**
(a) 5 km/h (b) 6 km/h
(c) 7 km/h (d) None of these
- 64.** In a race of 1000 m, A beats B by 100 m or 10 s. If they start a race of 1000 m simultaneously from the same point and if B gets injured after running 50 m less than half the race length and due to which his speed gets halved, then by how much time will A beat B? **2016 I**
(a) 65 s (b) 60 s (c) 50 s (d) 45 s
- 65.** In a race A, B and C take part. A beats B by 30 m, B beats C by 20 m and A beats C by 48 m.
Which of the following is/are correct?
I. The length of the race is 300 m.
II. The speeds of A, B and C are in the ratio 50 : 45 : 42.
Select the correct answer using codes given below.
(a) Only I (b) Only II **2016 I**
(c) Both I and II (d) Neither I nor II

ANSWERS

1	a	2	b	3	c	4	d	5	b	6	d	7	a	8	d	9	a	10	a
11	b	12	a	13	c	14	d	15	a	16	b	17	c	18	b	19	c	20	d
21	a	22	b	23	b	24	d	25	b	26	a	27	d	28	a	29	c	30	a
31	a	32	d	33	c	34	d	35	b	36	c	37	b	38	c	39	b	40	b
41	b	42	d	43	b	44	a	45	c	46	a	47	b	48	a	49	c	50	a
51	a	52	b	53	c	54	b	55	d	56	a	57	b	58	a	59	b	60	d

HINTS AND SOLUTIONS

1. (a) Let speed be v km/h.

We know that,

Distance = Speed $\times$ Time

$$6 \times 50 = 5 \times v$$

$$\Rightarrow v = \frac{6 \times 50}{5} = 60 \text{ km/h}$$

2. (b) Average speed = $\frac{2xy}{x+y}$ [Rule 2]

$$= \frac{2 \times 80 \times 20}{80 + 20} = 32 \text{ km/h}$$

3. (c) Let Sarita's speed be u and distance is constant.

$$\text{Then, } u \times 3 = (u - 12) \frac{15}{4}$$

$$\left[\because 3 \text{ h} + 45 \text{ min} = \frac{15}{4} \text{ h} \right]$$

$$\Rightarrow u = \frac{45 \times 4}{3} = 60 \text{ km/h}$$

$\therefore$ Distance = Speed $\times$ Time

$$= 60 \times 3 = 180 \text{ km}$$

4. (d) Here, $x = 10$, $y = 12$, $t_1 = \frac{6}{60}$ h and

$$t_2 = \frac{9}{60} \text{ h}$$

$$\therefore \text{ Required distance} = \frac{xy(t_1 + t_2)}{y - x}$$

[Rule 4]

$$= \frac{10 \times 12 \times 15}{60 \times 2}$$

$$= 15 \text{ km}$$

5. (b) Total distance covered

$$= 50 + 40 + 90 = 180 \text{ km}$$

Total time taken

$$= \left(\frac{50}{25} + \frac{40}{20} + \frac{90}{15} \right) = 10 \text{ h}$$

$\therefore$ Average speed for the whole journey

$$= \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

$$= \frac{180}{10} = 18 \text{ km/h}$$

$$\therefore 18 \text{ km/h} = \frac{18 \times 5}{18} \text{ m/s} = 5 \text{ m/s}$$

6. (d) Actual speed of boy = $(p - q)$ km/h

$$\text{Time taken to cover 1 km} = \frac{1}{p - q}$$

$$\therefore \frac{1}{p - q} = r \Rightarrow \frac{1}{r} = p - q$$

7. (a) Length of train = Distance covered in 40 s at the rate of 36 km/h.

[Rule 6 (i)]

$\therefore$ Length of train

$$= 40 \times 36 \times \frac{5}{18} = 400 \text{ m}$$

8. (d) Distance travelled in 44 s = $2\pi r$

$$= 2 \times \frac{22}{7} \times 21 = 132 \text{ m}$$

$$\therefore \text{ Speed} = \frac{132}{44} = 3 \text{ m/s}$$

$$\left[\because \text{ speed} = \frac{\text{distance}}{\text{time}} \right]$$

$$\text{Time taken to travel 3 km} = \frac{3000}{3}$$

$$= 1000 \text{ s} = \frac{1000}{60} \text{ min} = 16 \text{ min } 40 \text{ s}$$

9. (a) Distance between car and scooter = 30 km

Relative speed

$$y = 60 - 50 = 10 \text{ km/h}$$

So, the time taken by scooter to overtake

$$\text{the car} = \frac{30}{10} = 3 \text{ h}$$

10. (a) Let the buses meet x h after 6:00 am.

Then, the distance covered by the two buses is 250 km.

$$\therefore 40x + 60x = 250$$

$$\Rightarrow x = \frac{250}{100} = 2.5 \text{ h} = 2 \text{ h and } 30 \text{ min}$$

So, they will meet at 8 : 30 am.

11. (b) Let the distance between P and Q be x km and let the two trains meet y h after 7 am.

Then, T_1 covers x km in 4h and T_2 covers

x km in $3\frac{1}{2}$ h.

$$\therefore \text{ Speed of train } T_1 = \frac{x}{4} \text{ km/h}$$

$$\text{and speed of train } T_2 = \frac{2x}{7} \text{ km/h}$$

According to the question,

$$\frac{x(y+2)}{4} + \frac{2xy}{7} = x$$

$$\Rightarrow \frac{(y+2)}{4} + \frac{2y}{7} = 1$$

$$\Rightarrow \frac{7(y+2) + 8y}{28} = 1$$

$$\Rightarrow 7y + 14 + 8y = 28$$

$$\therefore y = \frac{14}{15} \text{ h} = \frac{14}{15} \times 60 \text{ min} = 56 \text{ min}$$

So, trains will meet at 7 : 56 am.

12. (a) Let x km distance be covered in y h.

Then, speed of object in first case

$$= \frac{x}{y} \text{ km/h}$$

As, half of this distance is covered in double time.

Then, speed of object in second case

$$= \frac{x}{2} \div 2y = \frac{x}{2} \times \frac{1}{2y} = \frac{x}{4y} \text{ km/h}$$

$\therefore$ Ratio of first and second speeds

$$= \frac{x}{y} : \frac{x}{4y} = 1 : \frac{1}{4} = 4 : 1$$

13. (c) Let the speed of two trains be x km/h and y km/h, respectively. Then, the time taken by first train to cover 110 km = Time taken by second train to cover 100 km.

$$\text{Thus, } \frac{110}{x} = \frac{100}{y} \Rightarrow \frac{x}{y} = \frac{110}{100}$$

$$\therefore x : y = 11 : 10$$

14. (d) Distance travelled by thief car in one hour = 80 km

Distance travelled in one hour by police car = 100 km

So, police travels extra 20 km in 1 h.

So, to overtake thief, police car has to travel 5 km extra.

$$\therefore \text{ Time} = \frac{5}{20} = \frac{1}{4} \text{ h} = \frac{60}{4} \text{ min} = 15 \text{ min}$$

15. (a) Let the speed of car A = x km/h

Let the speed of car B = y km/h

Total distance covered by both the cars in 1h = 70 km

$$x(1) + y(1) = 70$$

$$x + y = 70 \quad \dots(i)$$

[in opposite direction]

Distance covered by both the cars in

$$7 \text{ h} = 70 \text{ km}$$

But in same direction

$$7x - 7y = 70 \Rightarrow x - y = 10 \quad \dots(ii)$$

added Eq. (i) and (ii), we get

$$2x = 80, x = 40 \text{ km/h, } y = 30 \text{ km/h}$$

16. (b) Let the speed of car be v and distance covered by car in one litre be A .

$$\therefore A \propto \frac{1}{v^2} \Rightarrow A = \frac{K}{v^2}$$

Then, $A = 25 \text{ km/h}$ and $v = 30 \text{ km/h}$

$$\text{So, } 25 = \frac{K}{(30)^2} \Rightarrow K = 900 \times 25$$

$$\Rightarrow K = 22500 \Rightarrow A = \frac{22500}{v^2}$$